

EE 115 Homework 8

- 1) Assume that the frequency response $H(f)$ of a filter is such that $|H(f_1)| = 0.1$, $|H(f_2)| = 0.2$ and $|H(f_3)| = 0.01$. Let $y(t)$ be the output of the filter driven by the input $x(t)$. Determine the powers (in dB) of the input and output for each of the following cases:
 - a) **(20 points)** $x(t) = 2 \cos(2\pi f_1 t) + 3 \sin(2\pi f_2 t)$.
 - b) **(20 points)** $x(t) = 2 \cos(2\pi f_2 t) + 3 \sin(2\pi f_3 t)$.
 - c) **(10 points)** $x(t) = 3 \cos(2\pi f_1 t) + 4 \sin(2\pi f_2 t) + 5 \cos(2\pi f_3 t + \pi/3)$.

- 2) Consider a superheterodyne (or superhet) receiver which consists of an antenna followed by an “image-reject” filter $H_1(f)$ with tunable center frequency f_{RF} , a mixer using a tunable local oscillator’s frequency $f_{LO} = f_{RF} + f_{IF}$, and a bandpass filter (BPF) $H_2(f)$ with center frequency $f_{IF} = 445\text{kHz}$. Assume that the antenna can pick up signals at all frequencies from 100kHz to 10MHz, the desired station is at frequency 990kHz, and the desired bandpass signal has the bandwidth 20kHz.
 - a) **(20 points)** What should be the value of f_{LO} ?
 - b) **(20 points)** To avoid any interference, what is the maximum allowable bandwidth of the image-reject filter?
 - c) **(10 points)** Assume that $H_2(f)$ is an ideal bandpass filter with the center frequency 445kHz and bandwidth 20kHz, and the mixer uses $2 \cos(2\pi f_{LO} t)$. If $|H_1(f_{RF})| = 1$, $|H_1(f_{RF} + 2f_{IF})| = 0.1$ and the antenna also picks up a signal $w(t) = 3 \cos(2\pi(f_{RF} + 2f_{IF})t)$, what is the power of the interference caused by $w(t)$ at the output of $H_2(f)$?